Exam Design

I. Hashing & Hash Tables

- Expected time with a dict/set
- Worst case time for hash table.
- Resolving collisions through chaining

II. Graph Theory

- Directed & undirected
- Neighbors, successors, pred.
- Paths & path lengths, shortest paths
- Degree
- Connected components
- Number of edges in tree
- Reachability

III. Graph Representations

- Adjacency matrix, list, dict-of-sets
- Time complexity of basic operations

IV BFS

- Predecessors & BFS tree
- Queue properties
- Aggregate Analysis
- Time complexity on special graphs

V. DFS

- Predecessors & DFS tree
- Recursion tree properties
- Aggregate analysis
- Time complexity on special graphs
- Start & finish times
- DAGs
- Topological Sort

VI. Bellman-Ford

- Loop invariant
- Early stopping
- Negative cycle detection
- Time complexity

VII Dijkstra

- No negative edge weights
- Time complexity - Queue properties